

CHERAW, SC, USA

WMO: 720605

Lat: 34.717N

Lon: 79.950W

Elev: 73

StdP: 100.45

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00197

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.0	(c) -4.1	(d) -17.9	(e) 0.8	(f) 0.2	(g) -15.0	(h) 1.0	(i) 0.8	(j) 12.8	(k) 12.1	(l) 9.7	(m) 12.3	(n) 1.2	(o) 260	(p) 0.384

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	35.0	23.6	33.8	23.5	32.7	23.4	25.9	31.6	25.4	30.8	24.9	29.9	2.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.2	27.5	23.9	19.0	27.3	23.0	18.0	26.2	80.2	31.8	77.8	31.1	76.1	29.2	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 8.2	(b) 6.8	(c) 5.5	(e) -9.7	(f) 36.9	(g) 2.4	(h) 1.2	(i) -11.5	(j) 37.8	(k) -12.8	(l) 38.5	(m) -14.2	(n) 39.2	(o) -15.9	(p) 40.1		
			DB		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.7	5.5	8.2	11.0	16.8	21.5	25.4	26.7	25.9	23.1	17.0	10.7	7.8	(6)
(7)		DBStd	8.55	5.54	5.85	5.01	4.22	3.53	2.45	1.93	2.17	3.00	4.23	4.32	5.16	(7)
(8)		HDD10.0	459	160	97	49	3	0	0	0	0	0	2	42	106	(8)
(9)		HDD18.3	1648	397	287	232	78	15	0	0	0	3	75	232	328	(9)
(10)		CDD10.0	2898	22	46	80	207	357	461	519	493	394	219	62	39	(10)
(11)		CDD18.3	1046	1	3	4	32	113	211	261	235	148	33	2	2	(11)
(12)		CDH23.3	9676	2	23	60	332	1001	2066	2519	2148	1199	309	13	4	(12)
(13)		CDH26.7	4118	0	1	7	76	361	957	1193	967	478	78	1	0	(13)
(14)	Wind	WSAvg	2.1	2.2	2.5	2.5	2.1	2.0	1.9	1.7	2.0	2.2	2.0	1.9	(14)	
(15)	Precipitation	PrecAvg	1169	101	92	105	73	81	118	134	129	98	85	70	76	(15)
(16)		PrecMax	1588	202	210	229	179	174	256	262	275	333	309	167	188	(16)
(17)		PrecMin	755	20	18	16	1	21	15	16	10	8	0	18	5	(17)
(18)		PrecStd	173	46	47	48	44	40	63	59	65	66	65	37	42	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.9	26.2	27.6	30.3	34.9	37.1	37.0	35.2	34.0	32.0	25.2	23.8	(19)
(20)			MCWB	16.6	17.6	16.0	18.3	22.5	23.7	24.0	23.9	22.8	24.5	17.7	19.6	(20)
(21)		2%	DB	19.8	23.0	25.0	28.8	32.3	34.7	35.0	33.9	32.6	28.6	22.7	22.0	(21)
(22)			MCWB	14.8	15.5	15.4	18.5	21.5	23.2	24.0	23.8	22.4	20.9	16.7	18.7	(22)
(23)		5%	DB	17.5	20.3	22.7	27.4	30.2	32.9	33.7	32.8	31.3	27.1	21.0	19.0	(23)
(24)			MCWB	13.6	14.4	14.0	18.0	21.1	23.0	24.0	24.0	22.2	19.6	15.5	17.4	(24)
(25)		10%	DB	14.9	17.9	20.1	25.1	28.6	32.1	32.5	32.1	29.8	24.8	18.8	17.3	(25)
(26)			MCWB	11.4	13.3	13.2	16.9	20.8	22.8	23.8	23.9	21.9	18.9	14.9	15.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.8	19.4	18.9	22.1	24.3	25.7	27.0	26.6	25.5	24.8	20.3	21.3	(27)
(28)			MCDWB	20.8	23.1	22.4	25.7	29.5	32.4	33.5	31.8	30.7	28.5	22.2	22.4	(28)
(29)		2%	WB	16.9	18.0	17.4	21.0	23.3	25.1	26.1	25.9	24.6	23.5	18.9	19.5	(29)
(30)			MCDWB	18.4	21.0	20.1	24.6	28.2	31.5	32.1	31.1	29.3	26.4	21.0	20.8	(30)
(31)		5%	WB	14.7	16.5	16.4	20.0	22.6	24.5	25.5	25.4	23.9	22.1	17.5	17.7	(31)
(32)			MCDWB	16.4	20.3	20.5	24.1	27.4	30.3	30.9	30.4	27.9	25.0	18.7	18.1	(32)
(33)		10%	WB	12.2	14.4	15.1	18.8	21.9	23.9	24.9	24.8	23.1	20.2	15.9	15.2	(33)
(34)			MCDWB	14.1	17.6	19.3	22.9	26.9	29.2	29.8	29.4	26.4	23.2	18.6	16.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.6	12.2	12.8	13.5	12.0	12.0	11.4	10.9	11.4	12.8	13.2	10.9	(35)
(36)			MCDBR	14.4	14.6	16.1	15.4	14.3	13.5	12.7	12.3	13.7	14.7	14.8	12.1	(36)
(37)			MCWBR	9.7	8.6	8.1	6.6	5.1	4.4	4.1	4.0	4.9	6.9	9.3	8.7	(37)
(38)		5% WB	MCDWB	11.5	12.9	12.9	11.7	11.1	12.0	11.0	10.4	11.1	10.2	10.8	10.7	(38)
(39)			MCWBR	9.5	8.9	8.1	6.4	4.7	4.5	4.1	3.9	4.5	6.2	8.3	9.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.316	0.326	0.362	0.405	0.467	0.491	0.543	0.515	0.432	0.389	0.337	0.312	(40)	
(41)		taud	2.503	2.469	2.399	2.268	2.146	2.119	1.998	2.065	2.287	2.374	2.477	2.547	(41)	
(42)		Ebn at Noon	893	921	910	881	828	804	760	774	827	836	858	876	(42)	
(43)		Edh at Noon	83	96	112	134	153	158	177	162	124	105	85	76	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.70	3.34	4.43	5.55	6.09	6.41	6.06	5.51	4.70	3.87	2.96	2.34	(44)	
(45)		RadStd	0.15	0.37	0.34	0.40	0.49	0.44	0.37	0.26	0.40	0.48	0.24	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (73 neighbors)		+0.52	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+170	+124

Nomenclature: See separate page